

Wiring Diagram

ESP32 Pins

ESP32 Pin	Connects To	Purpose
3V3	Breadboard 3.3V rail, potentiometer outer leg	Power reference
GND	Breadboard ground rail	Shared ground
GPIO34	Potentiometer middle wiper	Analog input
GPIO25	330 ohm resistor -> LED anode	LED PWM output
GPIO26	1K resistor -> piezo buzzer +	Buzzer PWM output
GPIO27	Push button signal side	Mute button input

Component Returns

Component	Return Connection
Potentiometer low outer leg	GND rail
LED cathode	GND rail
Piezo buzzer -	GND rail
Push button other side	GND rail

The button uses the ESP32 internal pull-up resistor in firmware. Pressing the button connects GPIO27 to ground.